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SARDAR PATEL UNIVERSITY
M.Sc.-IT Master of Science in Information Technology
 Semester – I (NC) External Examinations, November 2017
PS01CINT03 – Introduction to Theoretical Computer Science (NC)
 Monday, 13th November, 2017

Time: 02:00 p.m. to 05:00 p.m.

Max Marks: 70

Q1. Choose the most appropriate option for each question.

[8]

- i. The set $A = \{a, b, c\}$ is _____ set.
 A) uncountable B) infinite
 C) finite D) None of these
- ii. A relation which is reflexive, transitive and symmetric is called as _____.
 A) equivalence relation B) poset
 C) angular relation D) None of these
- iii. Self loop is an example of _____.
 A) path B) circuit
 C) tree D) None of these
- iv. The degree of a vertex V having two self loops is _____.
 A) 1 B) 2
 C) 4 D) None of these
- v. A problem is considered as _____ if it can be solved by efficient algorithm.
 A) tractable B) intractable
 C) unsolvable D) None of these
- vi. A decline or changes that have occurred in ice-cream sales during november to february is called _____ variation.
 A) trend B) seasonal
 C) cyclic D) irregular
- vii. Which of the following equations represents the absorption law for a lattice?
 A) $a \vee (a \wedge b) = a$ B) $a \wedge (a \vee b) = b$
 C) Both A and B D) None of these
- viii. Fuzzy logic is _____ logic.
 A) multi valued B) single valued
 C) empty D) None of these

Q2. Answer the following questions (Any Seven):

[14]

- a. Define anti-symmetric relation and give one example of anti-symmetric relation.
- b. What is injective function?
- c. Define isomorphic graph and give one example of isomorphic graph.

- d. Show that in an undirected graph, total number of odd vertices is even.
- e. What is time complexity of an algorithm?
- f. Prove join and meet operations are associative in lattice.
- g. Give one example for (i) complete lattice and (ii) bounded lattice.
- h. Define secular trend. What are the uses of trend?
- i. Differentiate between fuzzy logic and crisp logic.

Q3. Answer the following questions:

- a. Define binary relation. Also explain reflexive, symmetric and anti-symmetric relations each with an example. [6]
- b. Write a note on types of grammars for different language sets. [6]

OR

- b. If $f(x)=x+7$ and $g(x)=x-3$ then find $f \circ g(x)$, $g \circ f(x)$, $f \circ g(4)$ and $g \circ f(4)$. [6]

Q4. Answer the following questions:

- a. Prove that there always exists a Hamiltonian path in a directed complete graph. [6]
- b. Explain Euler and Planner graphs with example. [6]

OR

- b. Write a note on complexity of problems. Explain by taking suitable example. [6]

Q5. Answer the following questions:

- a. Discuss cyclic variation, seasonal variation and irregular variation in time series. [6]
- b. Fit a Straight line trend by the method of Least Squares. Calculate the Trend Values. Estimate the value for 2019. [6]

Year	2004	2009	2010	2011	2012	2013	2014
Earnings (Rs. Lakhs)	70	75	90	91	95	98	100

OR

- b. Calculate 5-Yearly moving average for the following data to the numbers of commercial and industrial failures in a country during 1994 to 2009. [6]

Year	No. of Failures	Year	No. of Failures	Year	No. of Failures	Year	No. of Failures
1994	23	1998	20	2002	9	2006	12
1995	26	1999	12	2003	13	2007	9
1996	28	2000	12	2004	11	2008	3
1997	32	2001	10	2005	14	2009	1

Q6. Answer the following questions:

- a. Write a detailed note on fuzzy logic and its applications. [6]
- b. If fuzzy set $\tilde{A} = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0)\}$ and fuzzy set $\tilde{B} = \{(x_1, 0.7), (x_2, 0.2), (x_3, 1)\}$ then find $(\tilde{B} - \tilde{A}) \cup (\tilde{A} - \tilde{B})$. [6]

OR

- b. Given $T(\tilde{P}) = 0.55$ and $T(\tilde{Q}) = 0.8$, then find (i) $T(\tilde{P} \vee \tilde{Q})$, (ii) $T(\tilde{P} \Rightarrow \tilde{Q})$ [6]
